

Computational Docking Screen: Omeprazole Against *Escherichia coli* Resistance-Associated Proteins

An exploratory in silico screen - preliminary, hypothesis-generating, not validated

Status: Preliminary computational screen only. No wet-lab, cell-based, or functional data exists for this hypothesis at this time.

Purpose of this document: to report exactly what the docking simulation does and does not show, so the result can be cited accurately in any external or investor-facing context.

1. Summary

This report presents an exploratory molecular docking screen evaluating whether omeprazole, a widely used and approved proton pump inhibitor, shows predicted binding interaction with three class A beta-lactamases implicated in beta-lactam resistance: TEM-64, SHV-1 (R164H extended-spectrum mutant), and CTX-M-15. Ceftriaxone, a clinically active beta-lactam antibiotic subject to hydrolysis by these enzymes, was docked against the same targets as a reference comparator. Four additional compounds: verapamil, berberine, acetylcysteine, and bismuth subsalicylate were included as a broader comparator panel to contextualize omeprazole's binding profile against other repurposing candidates.

The screen found that omeprazole produces consistent, moderate-to-strong predicted binding scores across all three beta-lactamase targets, generally trailing ceftriaxone's affinity but remaining competitive at TEM-64 and CTX-M-15. This is a preliminary computational signal only. It identifies omeprazole as a candidate worth further investigation for beta-lactamase engagement, potentially as an inhibitor or "rescue" agent that restores beta-lactam efficacy but it does not demonstrate enzyme inhibition, restored antibiotic susceptibility, or any other biological effect.

2. Methods

Docking was performed using AutoDock Vina following standard receptor preparation in UCSF Chimera (removal of crystallographic waters and co-crystallized ligands, addition of polar hydrogens) and ligand preparation (generation of a 3D conformer with a standard protonation state). Grid box dimensions and center coordinates for each target were defined using DoGSiteScorer, with the co-crystallized ligand in each structure used as the reference point to anchor the binding site.

Three class A beta-lactamases were selected as targets based on their established roles in beta-lactam resistance:

TEM-64 beta-lactamase (PDB ID: 1JWZ): grid box centered on the active-site cleft surrounding the catalytic Ser70, using the co-crystallized boronic acid inhibitor as reference

SHV-1 R164H extended-spectrum beta-lactamase (PDB ID: 3OPR): grid box centered on the active-site cleft, using the co-crystallized SA2-13 ligand as reference, accounting for the R164H substitution associated with expanded-spectrum activity

CTX-M-15 extended-spectrum beta-lactamase (PDB ID: 4HBU): grid box centered on the active-site cleft, using the co-crystallized avibactam ligand as reference

The AcrB multidrug efflux transporter (PDB ID: 7OUK) was included as a fourth target, with the grid box centered on the transmembrane substrate-binding pocket using the co-crystallized BDM88855 inhibitor as reference, to provide broader context on whether candidate compounds engage a non-beta-lactamase resistance mechanism.

Ceftriaxone was docked against all four targets under the same conditions as a reference point, since it is a clinically active beta-lactam antibiotic subject to hydrolysis by these enzymes and provides a useful, though imperfect, benchmark for what a stronger predicted interaction looks like in this system. Omeprazole was the primary compound of interest, docked against all four targets to assess its potential as a beta-lactamase-engaging repurposing candidate. Verapamil, berberine, acetylcysteine, and bismuth subsalicylate were docked under identical conditions as a broader comparator panel to contextualize omeprazole's binding profile.

3. Results

3.1 Binding affinity scores for individual best poses (out of 9 models)

Target	Co-crystallized ligand	Drug Control: Ceftriaxone (kcal/mol)	Omeprazole (kcal/mol)	Verapamil (kcal/mol)	Berberine (kcal/mol)	Acetylcysteine (kcal/mol)	Bismuth subsalicylate (kcal/mol)
TEM-64 (1JWZ)	105: NA	-7.3	-7.3	-6.6	-7.5	-4.8	-6.1
SHV-1R 164H (3OPR)	SA2: -5.4	-7.4	-6.9	-6.6	-6.4	-4.5	-5.4
CTX-M-15 (4HBU)	NXL: -6.4	-8.1	-7.2	-6.4	-7.2	-4.6	-6.0
AcrB (7OUK)	1k8: -9.1	-6.0	-6.2	-6.3	-9.5	-4.3	-5.5

3.2 Molecular interactions of the raw proteins with their co-crystallized ligands

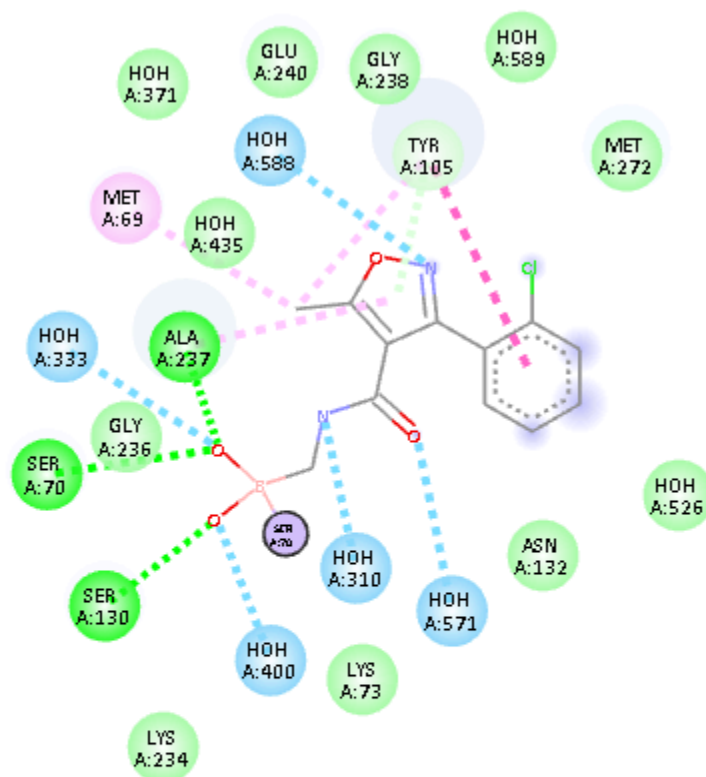


Fig 1. 2D protein-ligand interaction diagram of the raw TEM-64 (PDB: 1JWZ) of Escherichia coli with its co-crystallized ligand. Conventional hydrogen bonds with Ser70 and Ser130, a covalent bond at Ser70, several water-mediated hydrogen bonds, a Pi-donor hydrogen bond, Pi-Pi stacked and Pi-Pi T-shaped contacts with Tyr105, and an alkyl contact with Met69.

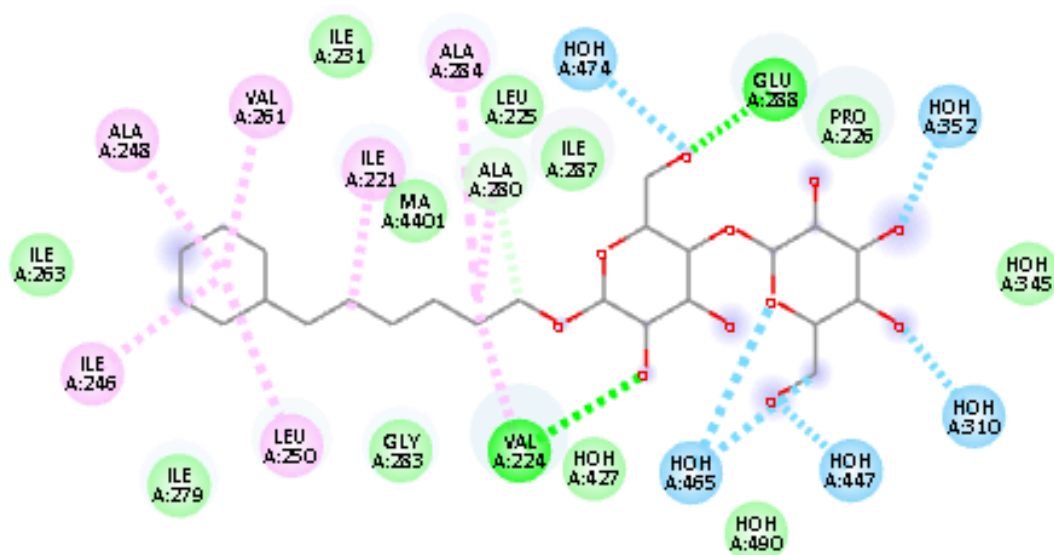


Fig 2. 2D protein-ligand interaction diagram of the raw SHV-1 R164H (PDB: 3OPR) of *Escherichia coli* with its co-crystallized ligand. The ligand engages a hydrophobic sub-pocket through van der Waals and alkyl contacts (Ala248, Val261, Ile263, Ile246, Ile279, Leu250, Ile231, Ala280, Ala284, Leu225, Ile221, Glu238, Pro226), a conventional hydrogen bond with Val224, and several water-mediated hydrogen bonds.

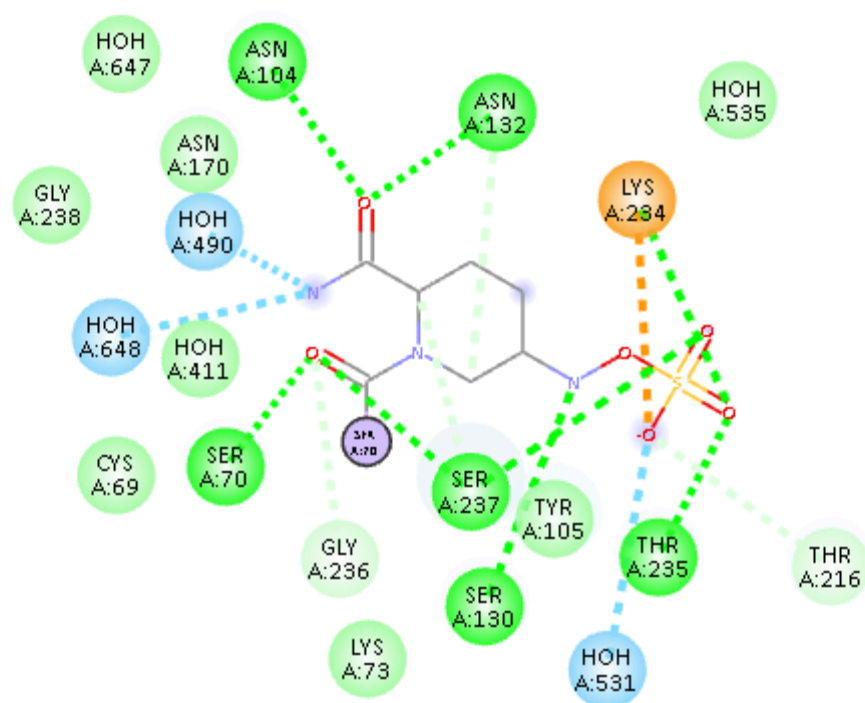


Fig 3. 2D protein-ligand interaction diagram of the raw CTX-M-15 (PDB: 4HBU) of *Escherichia coli* with its co-crystallized ligand. Conventional hydrogen bonds with Asn104, Asn132, Ser130, and Thr235, an attractive-charge interaction with Lys234, several water-mediated hydrogen bonds, and a covalent bond flagged at Ser70.

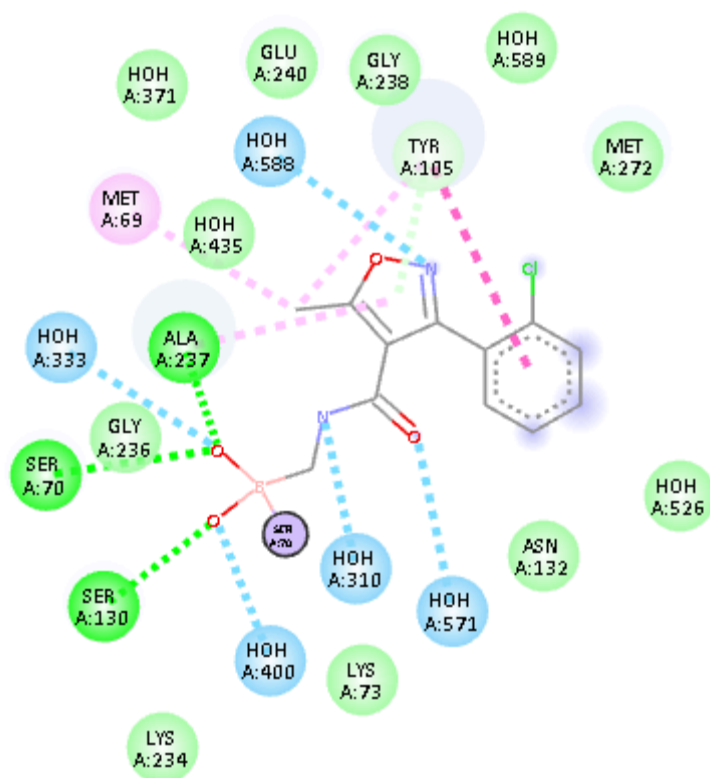


Fig 4. 2D protein-ligand interaction diagram of the raw ArcB (PDB: 7OUK) of *Escherichia coli* with its co-crystallized ligand. Entire engagement is hydrophobic/halogen-mediated: van der Waals contact across Ile943, Glu947, Phe948, Ala446, Asp407/408, Lys940, Leu449/404, a carbon-hydrogen-bond-type contact near its chlorine substituent to Leu442, Pi-sigma contacts with Val411 and Ile445, and alkyl/Pi-alkyl contacts with Ala441 and Ile438.

3.3 Molecular interactions of the six ligands with the proteins

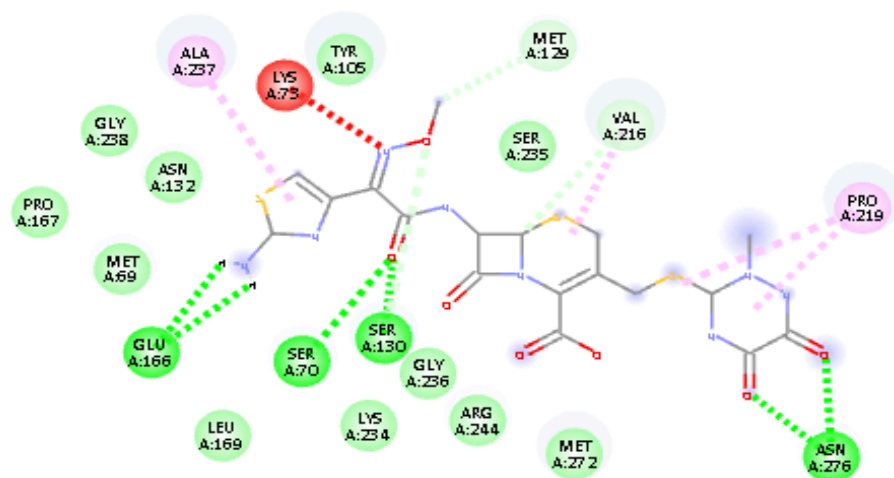


Fig 5. 2D protein-ligand interaction diagram of Ceftriaxone bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*, showing the most extensive and mechanistically complete interaction map: hydrogen bonds with Glu166 (the deacylation general base), Ser130, and Asn276, alongside a flagged unfavorable positive-positive clash with Lys75. Van der Waals and alkyl contacts blanket nearly the whole pocket (Ala237, Gly238, Asn132, Pro167, Met69/129/272, Leu169, Ser235, Lys234, Val216, Pro219, Arg244, Tyr105, Gly236).

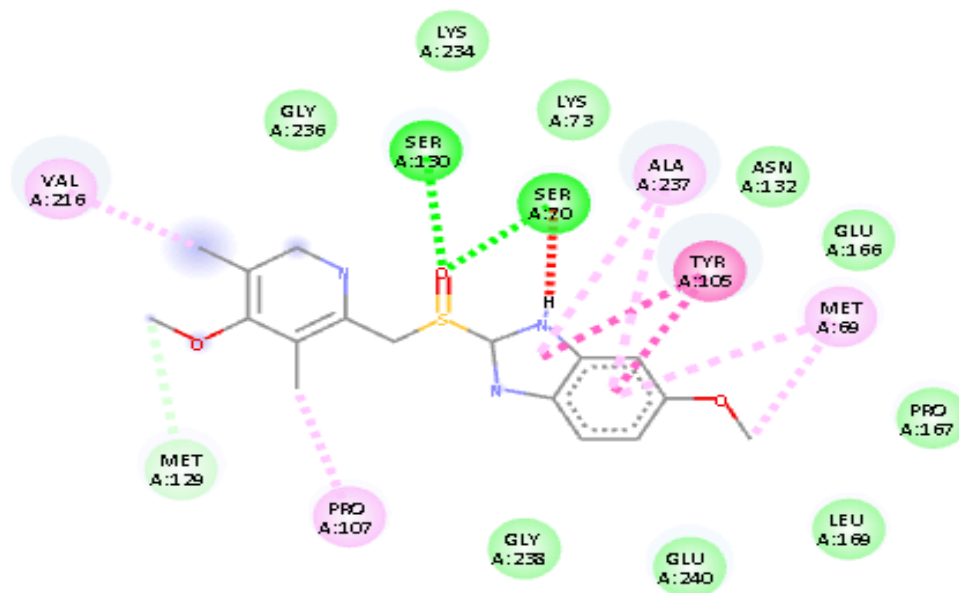


Fig 6. 2D protein-ligand interaction diagram of Omeprazole bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*. Omeprazole forms a conventional hydrogen bond with Ser70 and a carbon hydrogen bond with Ser130, plus a Pi-Pi T-shaped interaction with Tyr105. There's also a flagged "unfavorable donor-donor" contact near Ser70. Surrounding alkyl/Pi-Alkyl contacts with Met69, Pro107, Met129, Val216, and Ala237 fill out the hydrophobic walls of the pocket. This pose shows omeprazole occupying the same catalytic triad region (Ser70–Ser130–Tyr105) that the natural substrate would use.

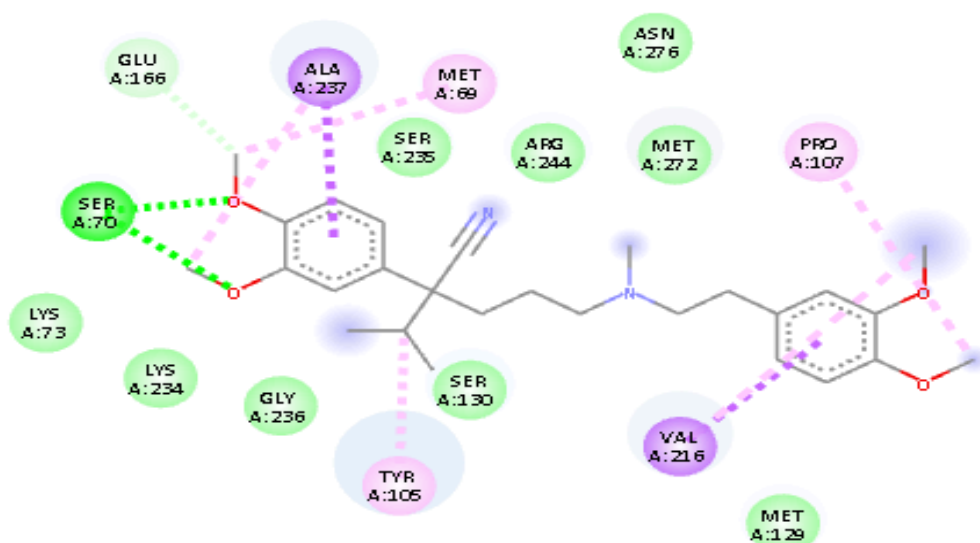


Fig 7. 2D interaction map of Verapamil bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*. It allows one aromatic end to hydrogen-bond with Ser70 (conventional) and Glu166 (carbon H-bond), while a Pi-Sigma contact with Ala237 and Alkyl/Pi-Alkyl contacts with Lys73, Lys234, Tyr105, Val216, Pro107, and Arg244 span the rest of the cleft.

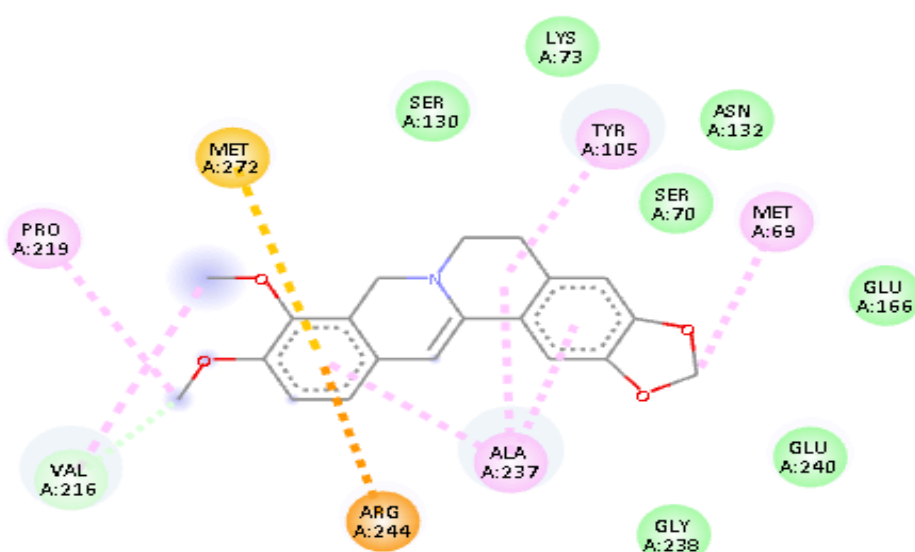


Fig 8. 2D protein-ligand interaction diagram of Berberine bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*. Berberine's fused polycyclic scaffold engages the pocket almost entirely through hydrophobic and charge-transfer-type interactions: a Pi-Sulfur contact with Met272, a Pi-Cation interaction with Arg244, and multiple alkyl/Pi-Alkyl contacts with Ala237, Met69, Tyr105, and Pro219, plus a carbon-hydrogen bond to Val216.

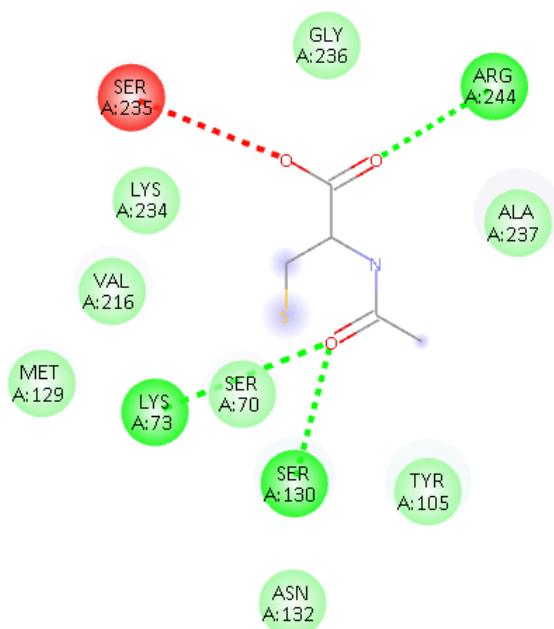


Fig 9. 2D protein-ligand interaction diagram of Acetylcystine bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*. Despite touching several key residues like hydrogen bonds with Arg244, Ser70, and Ser130, the interaction is undermined by an "unfavorable acceptor-acceptor" clash with Ser235, and the molecule's small size limits the total contact surface (mostly weak van der

Waals with Gly236, Lys234, Val216, Ala237). This is consistent with acetylcysteine's weak binding affinity across the whole panel.

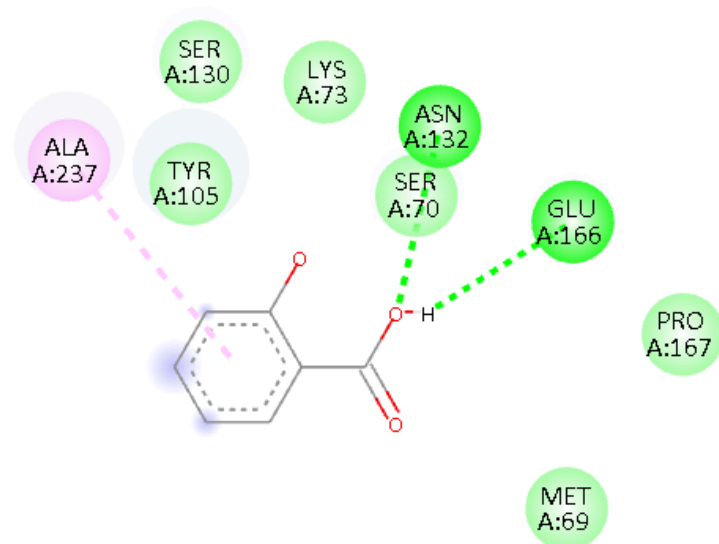


Fig 10. 2D protein-ligand interaction diagram of Bismuth Subsalicylate bound to TEM-64 (PDB: 1JWZ) of *Escherichia coli*. The salicylate hydroxyl/carboxyl group forms conventional hydrogen bonds with Ser70 and Glu166, placing this small molecule directly at the catalytic nucleophile and the deacylation-water-coordinating residue. A Pi-Alkyl contact with Ala237 anchors it within the oxyanion-hole region, with van der Waals support from Ser130, Lys73, Tyr105, and Met69.

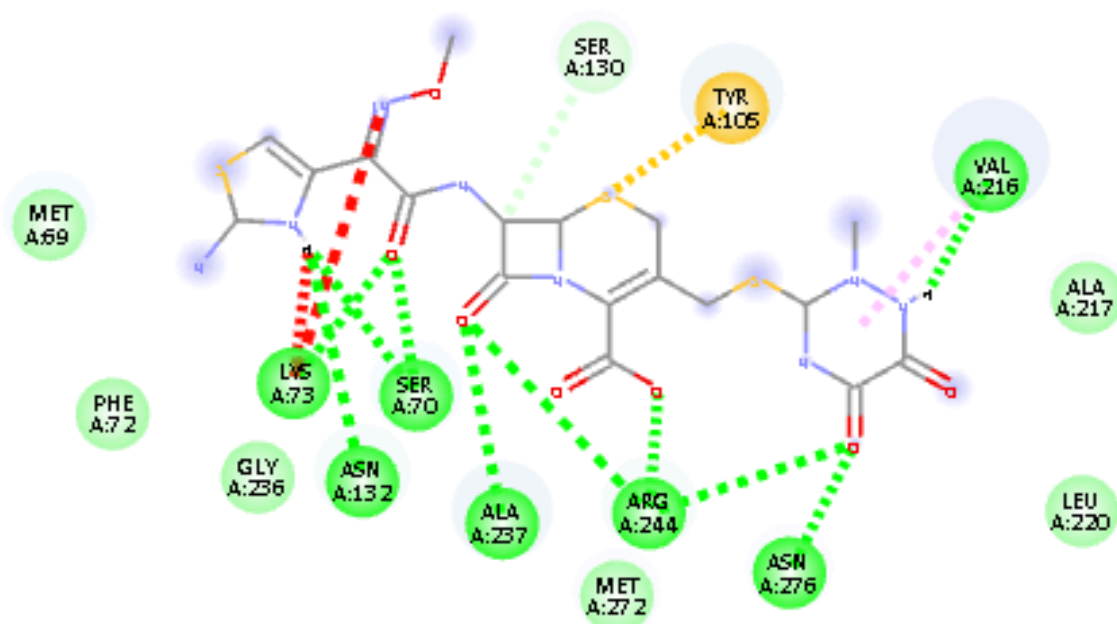


Fig 11. 2D protein-ligand interaction diagram of Ceftriaxone bound to SHV-1 R164H (PDB: 3OPR) of *Escherichia coli*. Hydrogen bonds with Ser70 (catalytic nucleophile), Asn132, Ala237, Arg244, and Asn276, a Pi-Sulfur interaction with Tyr105, and a flagged unfavorable donor-donor clash near Lys73.

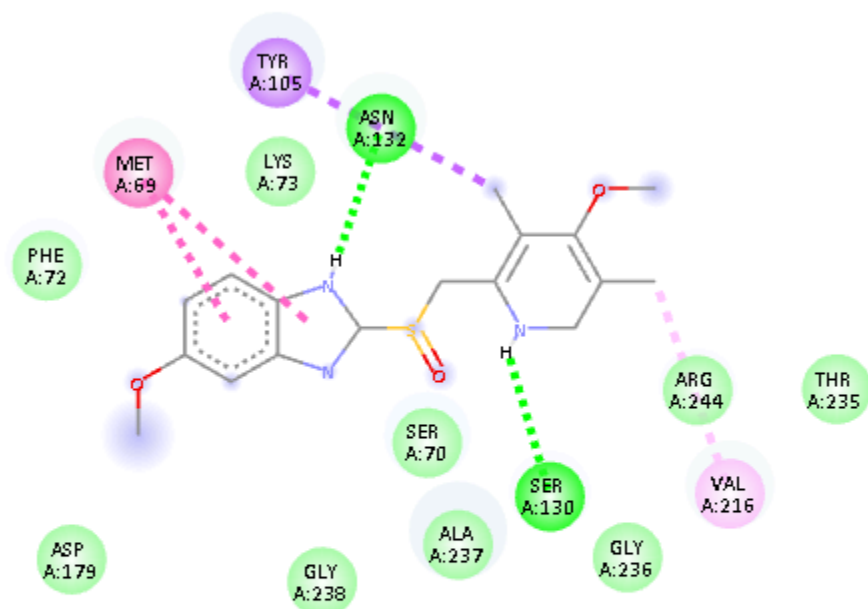


Fig 12. 2D protein-ligand interaction diagram of Omeprazole bound to SHV-1R164H (PDB: 3OPR) of *Escherichia coli*. Omeprazole here forms conventional hydrogen bonds with Asn132 and Ser130, with an amide-Pi stacked contact anchoring its benzimidazole ring near Met69, and

alkyl contacts with Arg244 and Val216. Ser70 is present only as a peripheral contact, not a direct hydrogen bond.

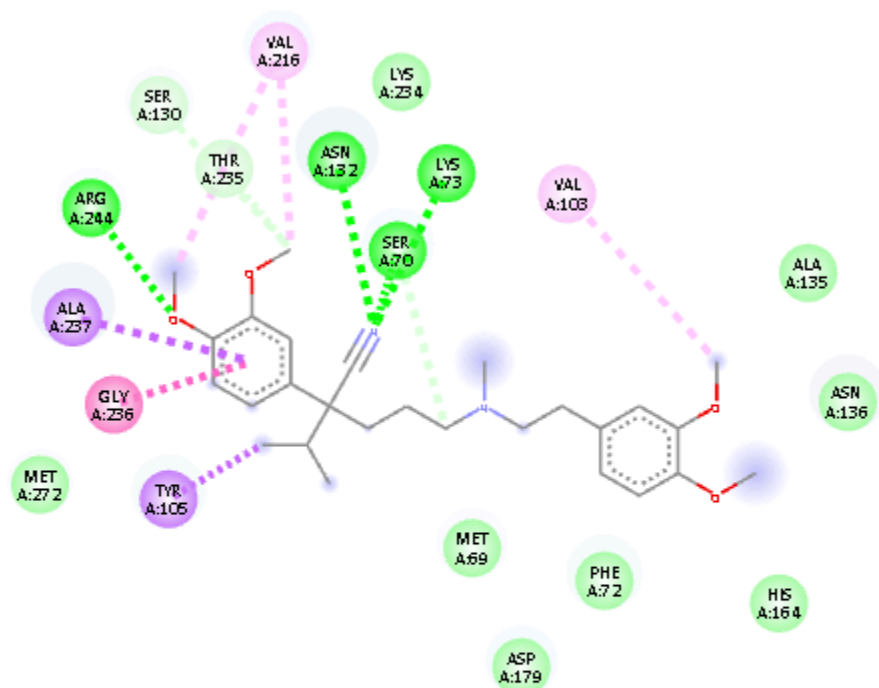


Fig 13. 2D protein-ligand interaction diagram of Verapamil bound to SHV-1 R164H (PDB: 3OPR) of *Escherichia coli*. Conventional hydrogen bonds to Arg244 and Asn132, a Pi-Sigma contact with Ala237, an amide-Pi stacked interaction near Ser70, plus peripheral alkyl/van der Waals contacts reaching out to Met69, Phe72, Asp179, and His164.

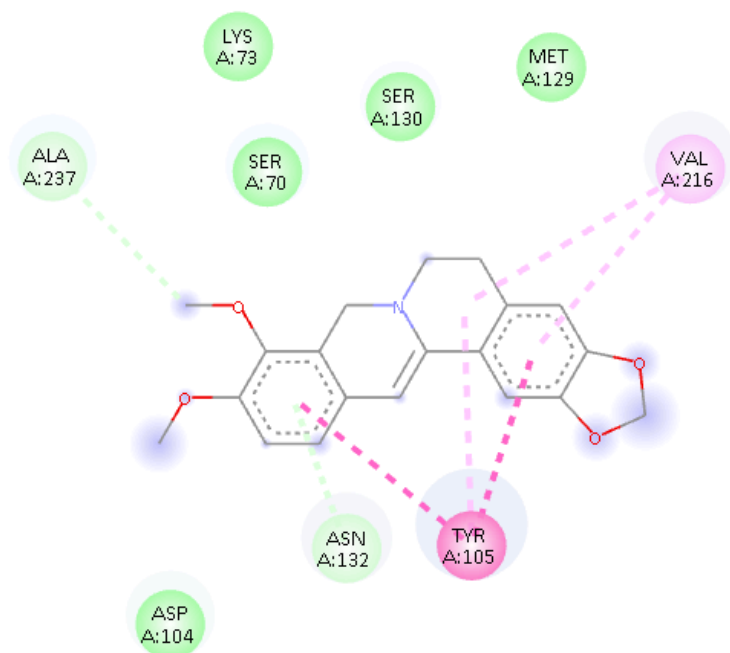


Fig 14. 2D protein-ligand interaction diagram of Berberine bound to SHV-1R164H (PDB: 3OPR) of Escherichia coli. Berberine engages this pocket almost entirely through hydrophobic and aromatic stacking, a Pi-Pi stacked contact with Tyr105, an alkyl contact with Val216, a carbon hydrogen bond to Ala237, and a weak pi-donor hydrogen bond to Asn132. Ser70 and Ser130 are only van der Waals contacts, not hydrogen bonds.

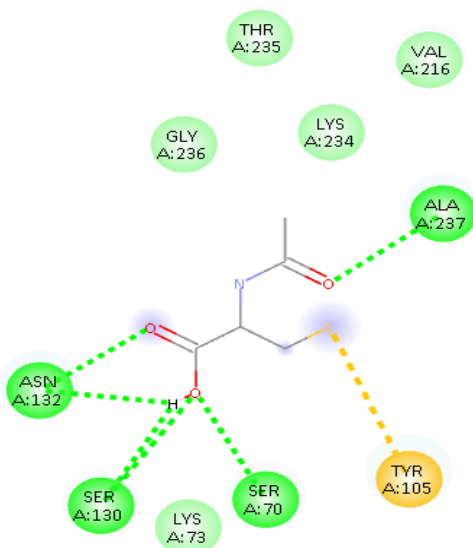


Fig 15. 2D protein-ligand interaction diagram of Acetylcysteine bound to SHV-1R164H (PDB: 3OPR) of Escherichia coli. It has four conventional hydrogen bonds (Ala237, Asn132, Ser130, and Ser70) plus a Pi-Sulfur contact from its thiol group to Tyr105

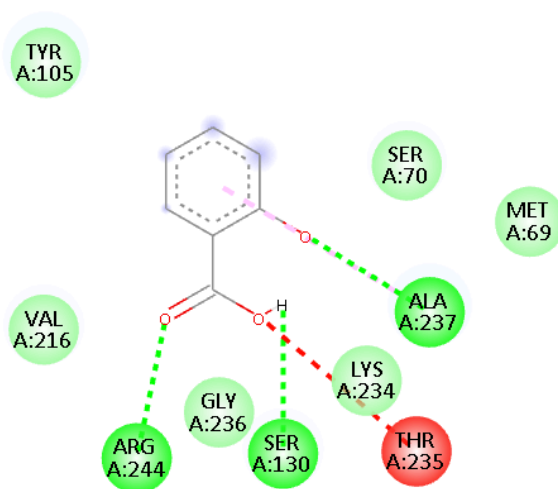


Fig 16. 2D protein-ligand interaction diagram of Bismuth subsalicylate bound to SHV-1R164H (PDB: 3OPR) of Escherichia coli. Bismuth's salicylate group forms conventional hydrogen bonds with Ala237, Arg244, and Ser130. Ser70 itself is only a van der Waals contact in this pose. There's also a flagged unfavorable acceptor-acceptor clash with Thr235.

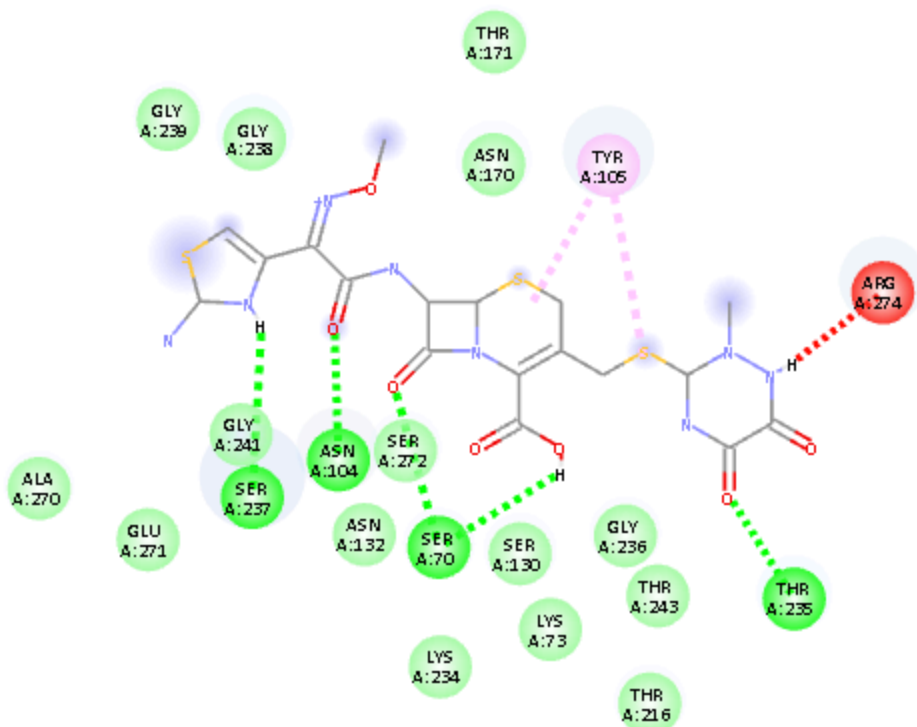


Fig 17. 2D protein-ligand interaction diagram of Ceftriaxone bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. Hydrogen bonds to Gly241, Asn104, Ser70, and Thr235, a Pi-alkyl contact with Tyr105, and a flagged unfavorable donor-donor clash with Arg274.

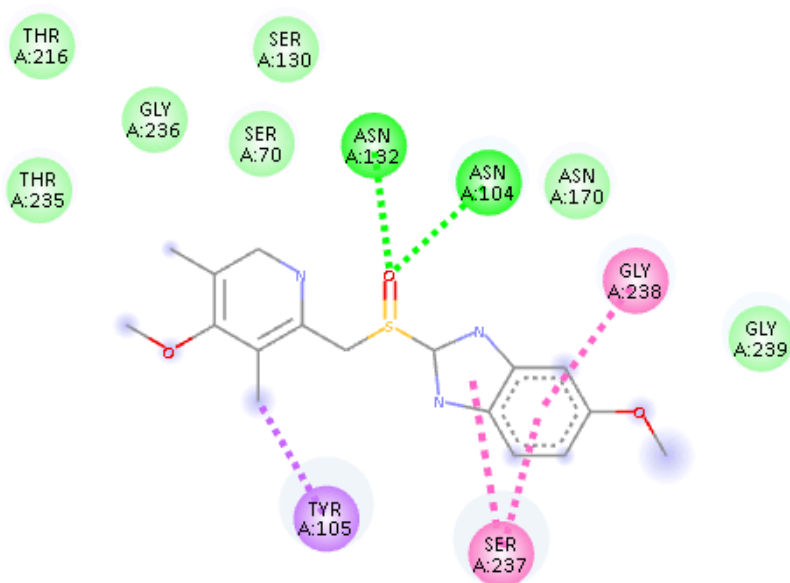


Fig 18. 2D protein-ligand interaction diagram of Omeprazole bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. Omeprazole forms conventional hydrogen bonds with Asn132 and Asn104, a

Pi-sigma contact with Tyr105, and amide-Pi stacked interactions near Gly238/Ser237. It does not directly hydrogen-bond Ser70 in this pocket (Ser70 shows up only as van der Waals).

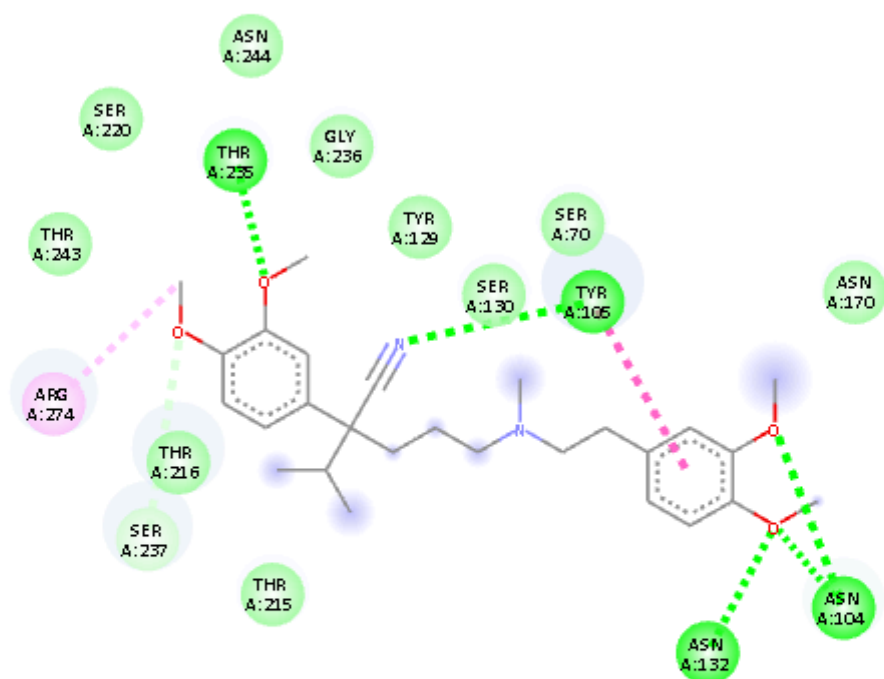


Fig 19. 2D protein-ligand interaction diagram of Verapamil bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. A conventional hydrogen bond to Thr235, a carbon hydrogen bond and Pi-Pi stacked contact both involving Tyr105, an alkyl contact with Arg274, and additional hydrogen-bond-type contacts near Asn104/Asn132.

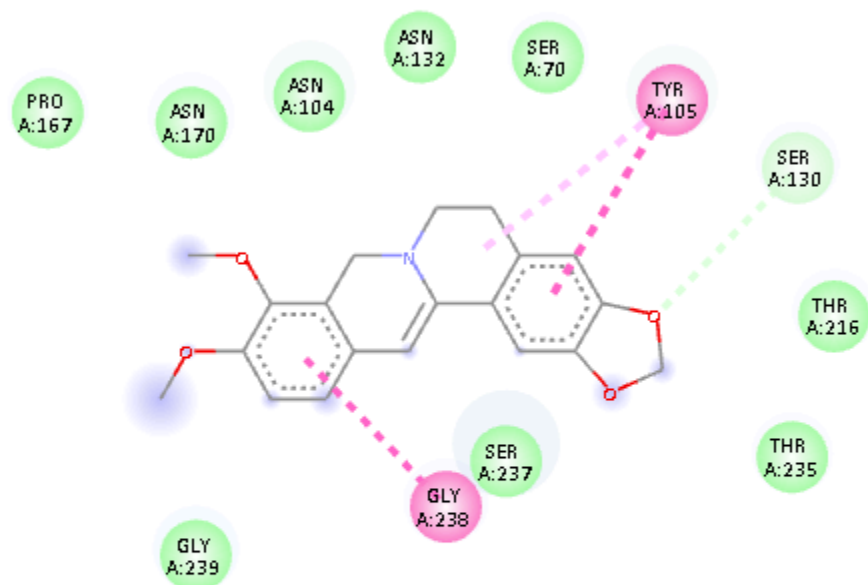


Fig 20. 2D protein-ligand interaction diagram of Berberine bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. Berberine here binds through aromatic/hydrophobic contacts only, an amide-Pi stacked interaction with Tyr105, a Pi-Pi T-shaped contact with Gly238, and a Pi-alkyl contact, with a carbon hydrogen bond to Ser130 as the only polar contact of note.

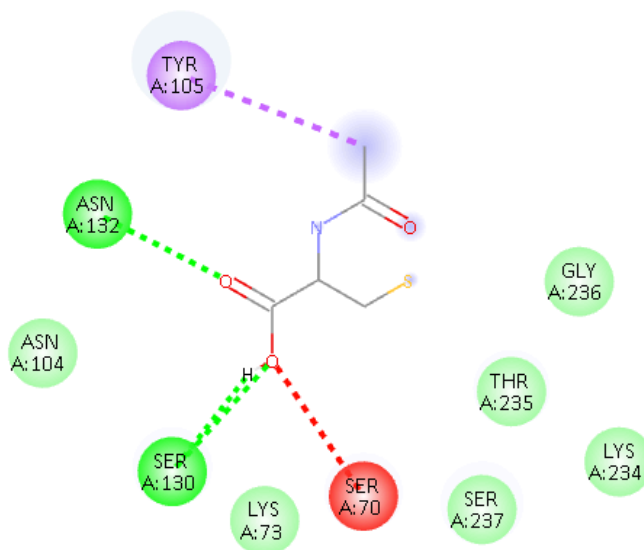


Fig 21. 2D protein-ligand interaction diagram of Acetylcystine bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. Two conventional hydrogen bonds (Asn132, Ser130) and a Pi-sigma contact with Tyr105, but an unfavorable acceptor-acceptor clash at Ser70.

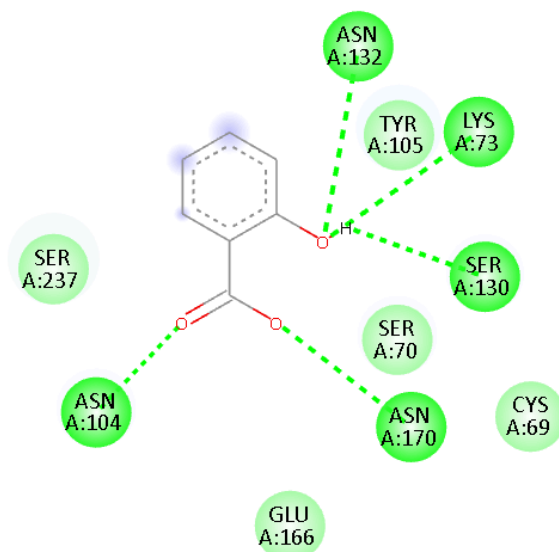


Fig 22. 2D protein-ligand interaction diagram of Bismuth subsalicylate bound to CTX-M-15 (PDB: 4HBU) of *Escherichia coli*. Four conventional hydrogen bonds, to Asn132, Ser130, Asn104, and Asn170. Ser70 and Glu166 appear only as van der Waals contacts.

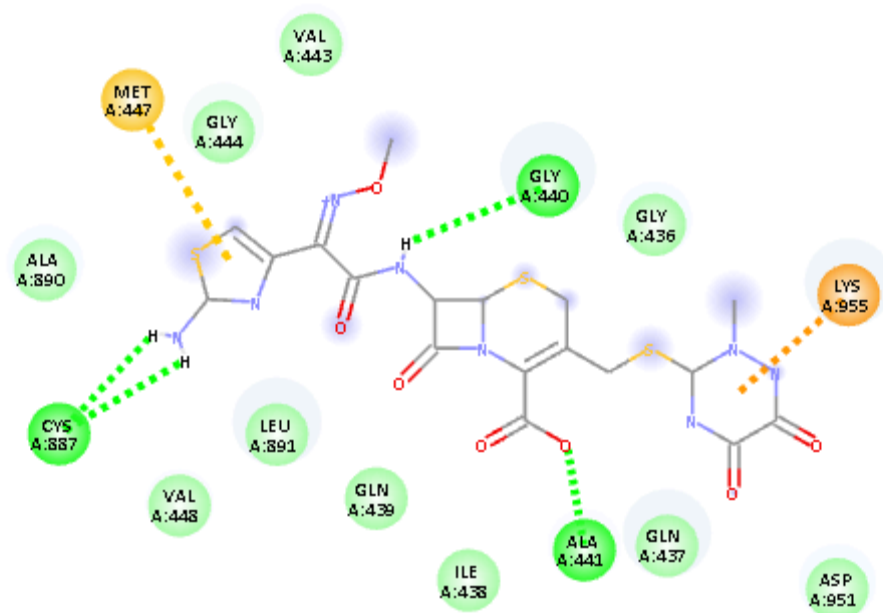


Fig 23. 2D protein-ligand interaction diagram of Ceftriaxone bound to ArcB (PDB: 7OUK) of *Escherichia coli*. Hydrogen bonds to Cys887 and Gly440, a Pi-sulfur contact with Met447, a Pi-cation contact with Lys955, plus alkyl contacts spanning Val443, Gly444/436, Leu891, Gln439/437, Ile438, Asp951.

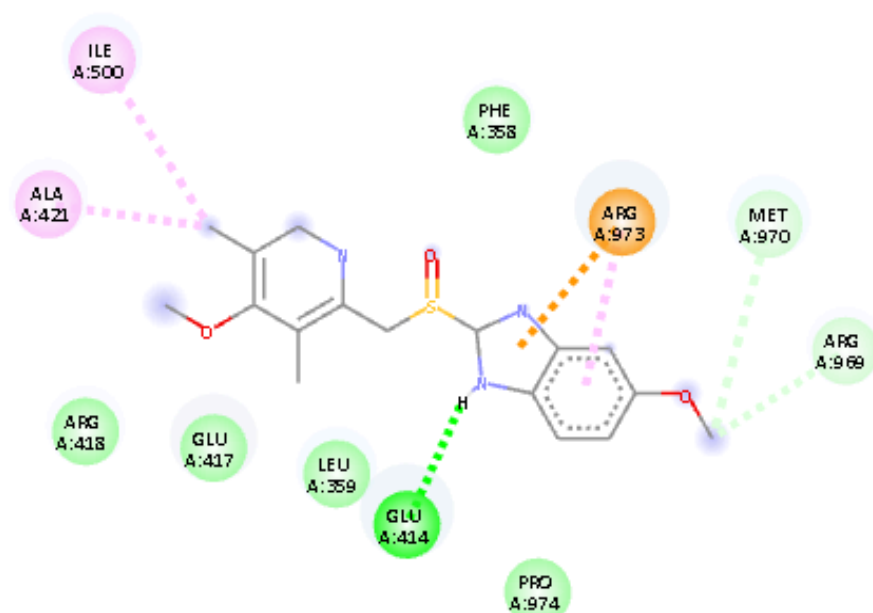


Fig 24. 2D protein-ligand interaction diagram of Omeprazole bound to ArcB (PDB: 7OUK) of Escherichia coli. A conventional hydrogen bond to Glu414, a Pi-cation contact with Arg973, and Pi-sigma/alkyl contacts with Met970/Arg969/Ile500/Ala421.

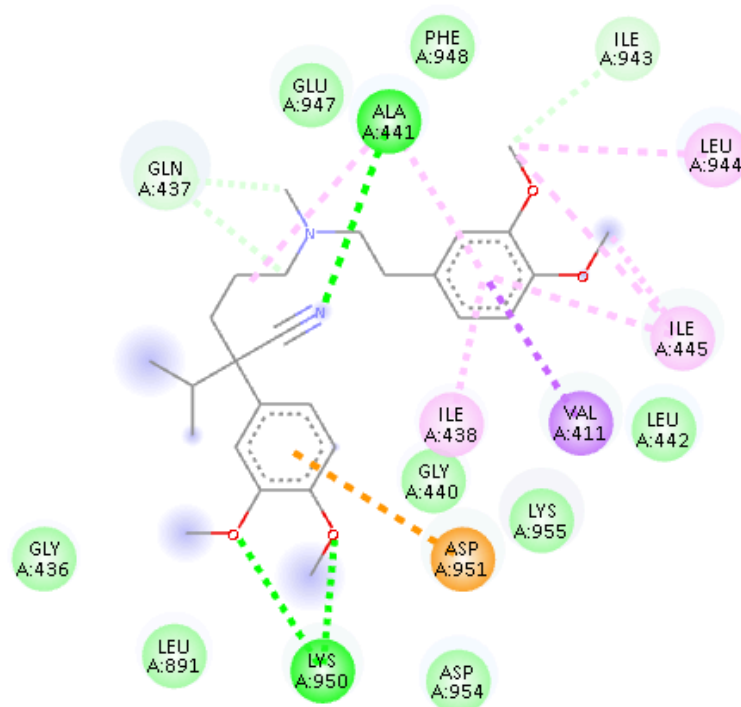


Fig 25. 2D protein-ligand interaction diagram of Verapamil bound to ArcB (PDB: 7OUK) of Escherichia coli. A conventional hydrogen bond to Ala441, a Pi-anion contact with Asp951, a Pi-sigma contact with Val411, and alkyl/Pi-alkyl contacts spanning Ile943/445/438, Leu944/442, Lys950/955.

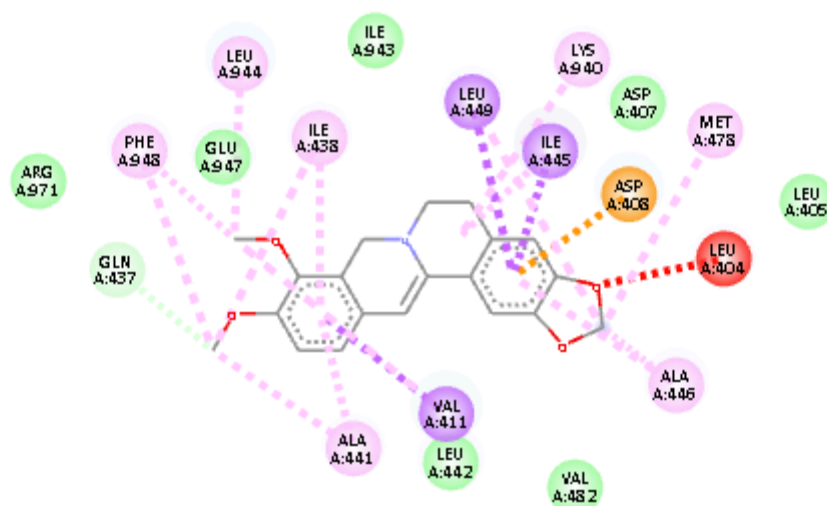


Fig 26. 2D protein-ligand interaction diagram of Berberine bound to ArcB (PDB: 7OUK) of *Escherichia coli*. It makes a Pi-anion contact with Asp408, dense alkyl/Pi-alkyl packing across Leu449, Ile445, Ile438, Ala441/446, Val411/482, Phe948, plus carbon hydrogen bonds to Gln437 and Glu947. There's one flagged unfavorable acceptor-acceptor contact near Leu404, but it's outweighed by the sheer breadth of favorable hydrophobic contacts.

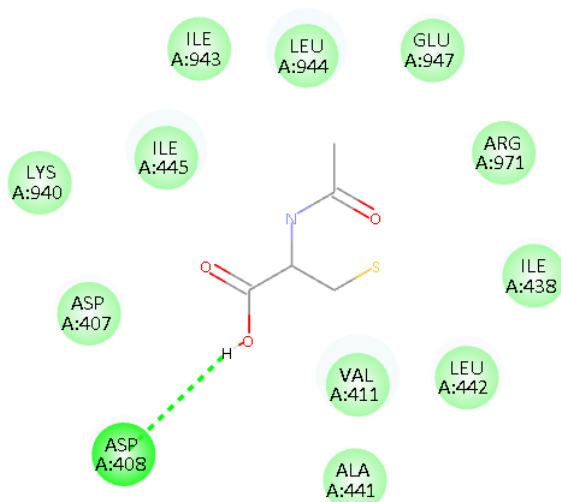


Fig 27. 2D protein-ligand interaction diagram of Acetylcysteine bound to ArcB (PDB: 7OUK) of *Escherichia coli*. Only a single conventional hydrogen bond (Asp408), with the rest of the contacts being weak van der Waals across Ile943/445/438, Leu944/442, Glu947, Lys940, Arg971, Val411, Ala441.

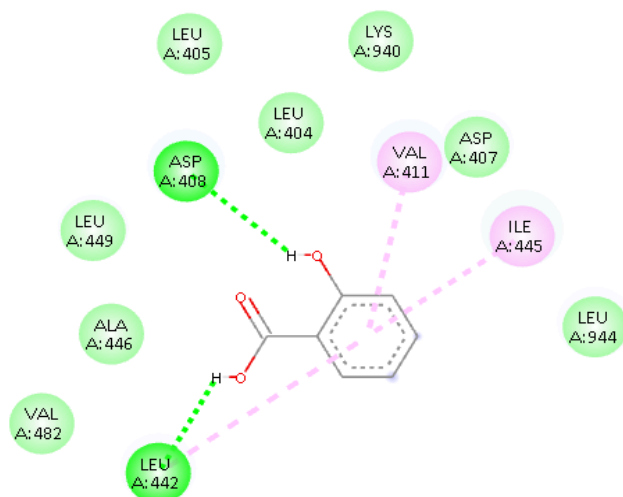


Fig 28. 2D protein-ligand interaction diagram of Bismuth subsalicylate bound to ArcB (PDB: 7OUK) of *Escherichia coli*. Two conventional hydrogen bonds (Asp408, Leu442) plus modest van der Waals contact with the surrounding hydrophobic shell (Leu405/449/944, Val411/482, Ile445, Ala446).

Table1. Molecular docking interactions of proposed drugs with **TEM-64 (1JWZ)**, **SHV-1(3OPR)**, **CTX-M-15(4HBU)** and **ArcB(7OUK)** of *Escherichia coli*

Target (PDB)	Ligand	Hydrogen Bond Residues	Additional Interactions
TEM-64 (1JWZ)	Ceftriaxone	Glu166, Ser130, and Asn276	Ala237, Gly238, Asn132, Pro167, Met69/129/272, Leu169, Ser235, Lys234, Val216, Pro219, Arg244, Tyr105, Gly236
	Omeprazole	Ser70	Ser130, Tyr105, Met69, Pro107, Met129, Val216, and Ala237
	Verapamil	Ser70	Glu166, Ala237, Lys73, Lys234, Tyr105, Val216, Pro107, and Arg244

	Berberine	NA	Val216, Met272, Arg244, Ala237, Met69, Tyr105, and Pro219
	Acetylcystine	Arg244, Ser70, and Ser130	Gly236, Lys234, Val216, Ala237
	Bismuth subsalicylate	Ser70 and Glu166	Ala237, Ser130, Lys73, Tyr105, and Met69
SHV-1 R164H (3OPR)	Ceftriaxone	Ser70, Asn132, Ala237, Arg244, and Asn276	Tyr105
	Omeprazole	Asn132 and Ser130	Met69, Arg244 and Val216.
	Verapamil	Arg244 and Asn132	Ala237, Ser70, Met69, Phe72, Asp179, and His164
	Berberine	Asn132	Ala237, Tyr105, Val216, Ser70 and Ser130
	Acetylcystine	Ala237, Asn132, Ser130, and Ser70	Tyr105
	Bismuth subsalicylate	Ala237, Arg244, and Ser130	Ser70
CTX-M-15 (4HBU)	Ceftriaxone	Gly241, Asn104, Ser70, and Thr235	Tyr105
	Omeprazole	Asn132, Asn104, Gly238/Ser237	Tyr105 and Ser70
	Verapamil	Thr235, Asn104/Asn132	Tyr105 and Arg274
	Berberine	NA	Tyr105, Gly238, Ser130
	Acetylcystine	Asn132 and Ser130	Tyr105
	Bismuth subsalicylate	Asn132, Ser130, Asn104, and Asn170	Ser70 and Glu166
ArcB (7OUK)	Ceftriaxone	Cys887 and Gly440	Met447, Lys955, Val443, Gly444/436, Leu891, Gln439/437, Ile438, Asp951.
	Omeprazole	Glu414	Arg973, Met970/Arg969/Ile500/Ala421.

	Verapamil	Ala441	Asp951, Val411, Ile943/445/438, Leu944/442, Lys950/955.
	Berberine	NA	Asp408, Leu449, Ile445, Ile438, Ala441/446, Val411/482, Phe948, Gln437 and Glu947
	Acetylcystine	Asp408	Ile943/445/438, Leu944/442, Glu947, Lys940, Arg971, Val411, Ala441.
	Bismuth subsalicylate	Asp408 and Leu442	Leu405/449/944, Val411/482, Ile445, Ala446

Molecular docking analysis demonstrated that omeprazole interacted favorably with all three class A beta-lactamase targets: TEM-64 (PDB: 1JWZ), SHV-1 R164H (PDB: 3OPR), and CTX-M-15 (PDB: 4HBU) with binding affinities and interaction profiles that partially overlapped those of ceftriaxone, the clinically active reference antibiotic. Ceftriaxone exhibited the most extensive hydrogen-bonding networks across all three enzymes, engaging the catalytic Ser70 directly at TEM-64 and CTX-M-15, along with substrate-recognition-loop residues including Ser130, Asn104, Asn132, Ala237, Arg244, and Asn276, reflecting stable, catalytically-anchored binding consistent with its role as the natural substrate. Omeprazole formed a comparatively smaller but still substantial contact network, consistently engaging the substrate-positioning residues Ser130 and Asn132 across all three enzymes, with a direct hydrogen bond to the catalytic Ser70 specifically at TEM-64. At SHV-1 R164H and CTX-M-15, omeprazole's engagement of Ser70 was reduced to a peripheral contact, with binding instead anchored through Asn132, Asn104, and Ala237. Binding affinities for omeprazole (-6.9 to -7.3 kcal/mol) trailed ceftriaxone's (-7.3 to -8.1 kcal/mol) at all three targets but remained within a comparable range, particularly at TEM-64, where the two compounds' affinities were nearly identical.

Within the AcrB multidrug efflux transporter (PDB: 7OUK), omeprazole showed comparatively limited engagement of the hydrophobic substrate-binding channel, forming a single hydrogen bond with Glu414 alongside modest hydrophobic contacts, corresponding to a binding affinity (-6.0 to -6.2 kcal/mol) similar to ceftriaxone's (-6.0 kcal/mol) and considerably weaker than berberine's (-9.5 kcal/mol), which formed an extensive hydrophobic contact network within the same pocket.

3.4 Discussion

Across the three beta-lactamase targets, ceftriaxone produced a more favorable (more negative) predicted binding score than omeprazole at SHV-1 R164H and CTX-M-15, differences of roughly 0.5 and 0.9 kcal/mol respectively, consistent with ceftriaxone's role as the natural substrate of these enzymes. At TEM-64, however, omeprazole's score matched ceftriaxone's exactly (-7.3 kcal/mol), and at AcrB omeprazole's score was marginally stronger than ceftriaxone's. These are real but not large gaps in docking terms, and the pattern is consistent

with omeprazole having genuine and moderate predicted affinity for the beta-lactamase active sites. But it is equally consistent with a range of other explanations, including imprecision in the Vina scoring function, a predicted pose that differs from the true binding mode, or an interaction that is real but functionally irrelevant to catalysis. The docking data alone cannot distinguish between these possibilities.

Confidence in these results is target-dependent, reflecting differences in pocket definition. Grid boxes were derived using DoGSiteScorer based on each structure's co-crystallized ligand, with ligand coverage of the predicted pocket volume at 100.0% for CTX-M-15 and AcrB, 81.25% for TEM-64, and 55.81% for SHV-1 R164H. The lower coverage at SHV-1 R164H explains why that ligand's interaction profile did not overlap with the catalytic Ser70/Ser130/Tyr105 residues engaged by the docked compounds, even though the grid box itself did capture those residues. For AcrB, only the "L-promoter" pocket was used, as it was the sole cavity corresponding to a functionally engaged, ligand-accessible region of the transporter.

As an additional benchmark, each target's own co-crystallized ligand affinity was considered alongside ceftriaxone as a reference. Together, these two comparators show omeprazole's affinities were consistently weaker than or comparable to ceftriaxone's, with an interaction profile partially overlapping the catalytic engagement of each target's native ligand, most informatively at TEM-64 and CTX-M-15, where pocket coverage was highest, and least so at SHV-1 R164H, given its reference ligand's partial coverage.

4. What Can and Cannot Be Claimed From This Data

Supportable claim	Not supportable from this data
Omeprazole predicted binding at TEM-64 and CTX-M-15 is comparable in magnitude to ceftriaxone's, and consistently engages the same substrate-recognition loop (Ser130/Asn132) across all three beta-lactamases.	"Omeprazole inhibits beta-lactamase activity" or "omeprazole restores ceftriaxone efficacy."
Berberine's binding mode differs mechanistically from the other candidates: strong, broad hydrophobic/aromatic engagement at AcrB and TEM-64, with essentially no catalytic-serine hydrogen bonding at any beta-lactamase target.	"Berberine is a better drug candidate than omeprazole"

5. Conclusion

This docking screen provides a preliminary, exploratory computational signal suggesting omeprazole may interact with three beta-lactamase enzymes implicated in antimicrobial resistance at binding affinities comparable to, and in one case matching, the clinical comparator ceftriaxone. Its interaction pattern consistently engaged the substrate-recognition loop across all three enzymes and directly contacted the catalytic Ser70 at TEM-64, while showing limited engagement with the AcrB efflux transporter, supporting a beta-lactamase-focused mechanistic hypothesis. Berberine separately emerged as a strongly predicted AcrB binder, warranting consideration as an efflux-focused candidate outside this analysis's scope.

This result should be treated strictly as hypothesis-generating. Docking cannot confirm true binding affinity, functional enzyme inhibition, or restored antibiotic susceptibility, and wet-lab validations including enzyme inhibition assays, synergy testing with ceftriaxone, and ideally structural confirmation are required before any claim of beta-lactamase-inhibitory or antimicrobial activity can be supported. This framing is intentional and should be preserved in any future internal, academic, or investor-facing use of this work.

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